Comparative Economics

Week II – October 9th, 2026

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https://dl1.cuni.cz/course/view.php?id=453

26$Inst1tutions

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Outline of the Session

- Lecture session
  - Course organization
  - Problems with economic data under different economic systems
  - Empirical analysis of economic systems
    - Synthetic control method
- Seminar session
  - Focus on institutions
  - Inclusive v. extractive institutions.
  - Dynamics of economic systems – Acemoglu & Robinson
  - Role of property rights.
  - Role of the state.

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Reading Assignment for the Next Session (October 16th)

- Acemoglu & Robinson – chapter 1 from **The Narrow Corridor**
  - Explain Hobbes' "Warre" and its effects on economic behaviour.
  - What do Acemoglu and Robinson accept in Hobbes's argument, and what are their two principal objections?
  - What are the differences between the Absent, Shackled, and Despotic Leviathan?
  - What is the "cage of norms"?
  - How can the same social norms both protect people and restrict their liberty? Explain using an example from the chapter.
- Acemoglu & Robinson (2006): **Paths of Economic and Political Development**
  - What is the main driving force of the development?
  - Can you very shortly summarize their theoretical model, i.e. linkages between institutions and power?
  - What determines the de facto power?
  - Why does the credibility of commitment matter?
  - When does the elite choose a repression as a reaction to democratization of society?
  - Did there seem to be any relationship between the economic rise of the UK and its democratization?
- You **do not have to submit these answers in writing**, just be ready to answer (and discuss) them.

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Texts and Rules for the First "Summary"

- Texts that will be discussed in the first "summary":
  - Roland – chapter 7: Institutions and Economic Development
  - Acemoglu & Robinson (2006): Paths of Economic and Political Development

- Rules for the "summary":
  - You will write the summary during the last twenty minutes of the third seminar session (October 16th).
  - There will be several multiple-choice questions and 1-2 open-ended question related to the aforementioned two texts.

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Data and Empirical Analysis

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Problems with Positive Analysis of Economic Systems

- A limited number (representativeness) of units
  - Are the specific features observed in a particular country related to its history/culture or economic model?
- Data with qualitative features needed (institutions!)
- Endogeneities
- Comparability of data
- Methods of comparisons
  - Ideally we would need to know how the same country would have evolved under a different arrangement
    - Counterfactuals

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Comparability of data

- Time dimension
- Statistical methodologies
- Reliability and logic of the data-collection process
  - Distortions - e.g. due to the logic of the functioning of the domestic political regime
- Availability of data for some economies
  - Censorship
  - Self-censorship (e.g. effects on surveys)

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Data Issues….

Who controls the past controls the future,
who controls the present controls the past.

"China's gross domestic product (GDP) figure is man-made and therefore unreliable"

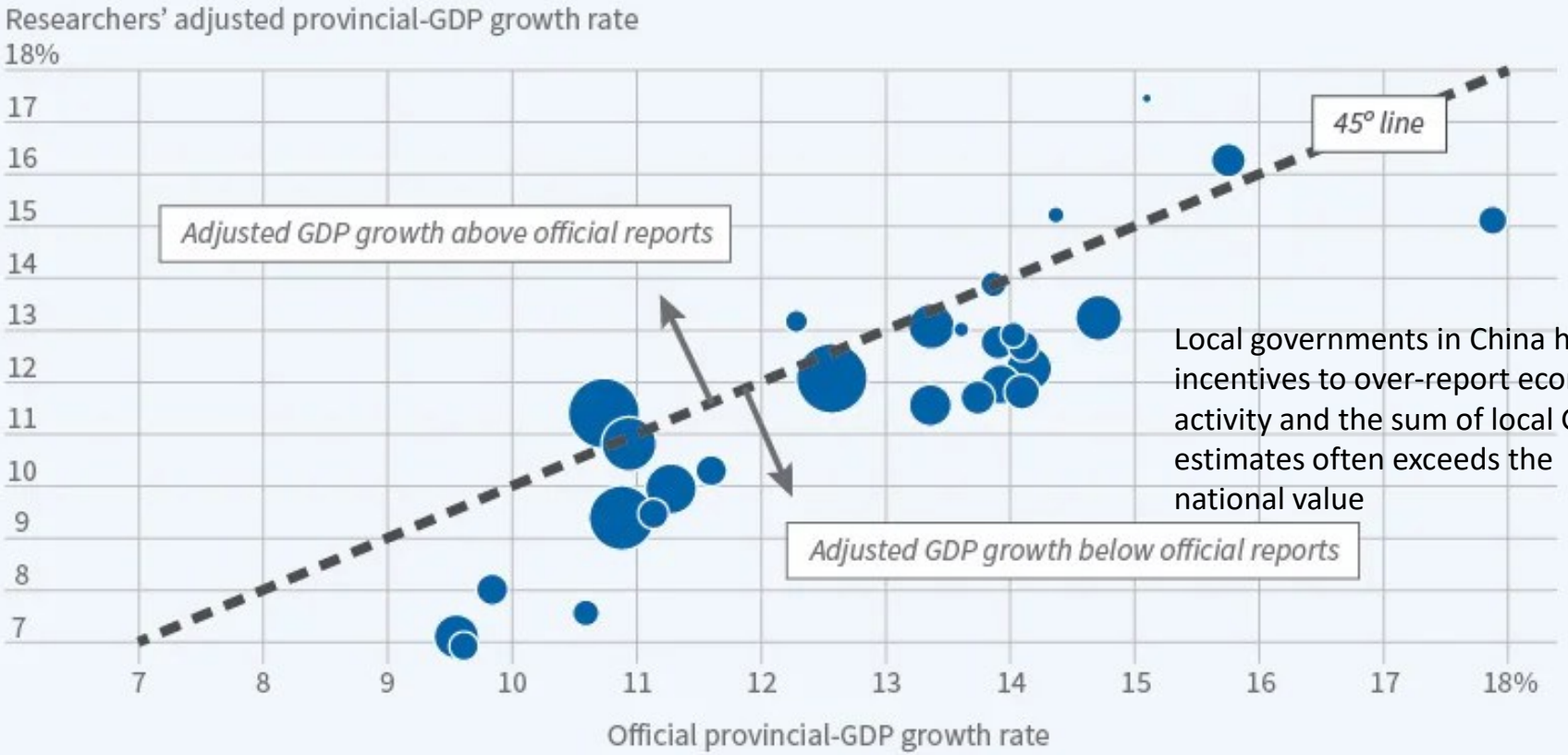
Li Keqiang, Chinese PM 2013-23

Suggested to look at other indicators: electricity consumption, volume of rail cargo, and bank loans



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GDP of Chinese Provinces: Officially Reported vs. Researchers' Estimates



Local governments in China have incentives to over-report economic activity and the sum of local GDP estimates often exceeds the national value

Circle size represents relative size of provincial GDP

Source: Researchers' calculations using data from China's National Bureau of Statistics and State Administration of Taxation

Source: Chen et al (2019)

Hide the Troublesome Indicators

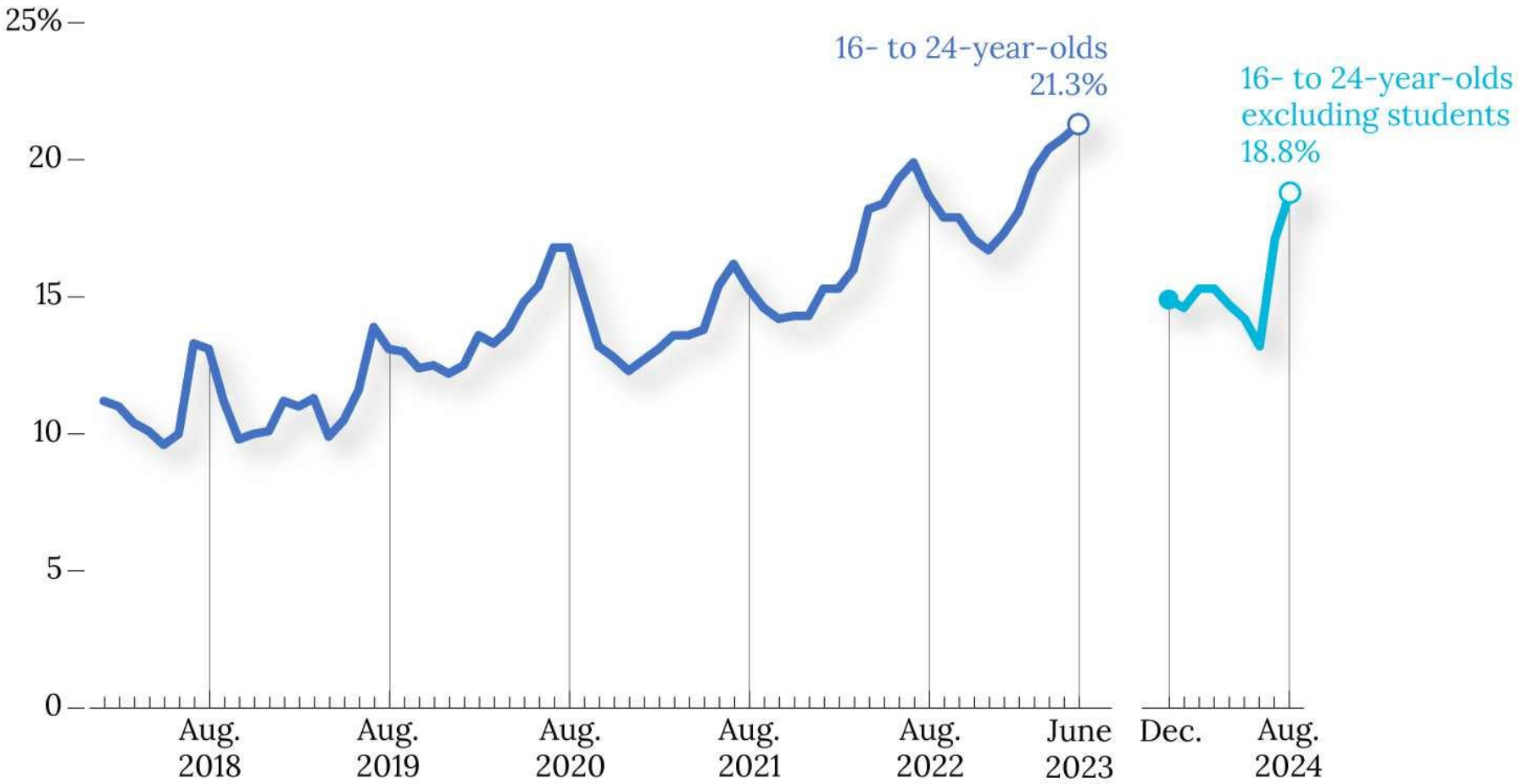
- Faking data can be difficult
  - It can be easier to hide troublesome indicators
  - Combination possible: hiding data can help you obscure possible problems in the published data

- Feng & Douglas (2025):
  - Chinese officials have stopped publishing hundreds of data points once used by researchers and investors
  - Examples:
    - Land sales measures, foreign investment data and unemployment indicators,…

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China's Youth Jobless Rate Hits Record for Second Month

Surveyed urban unemployment rate



Sources: National Bureau of Statistics, CEIC

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National land sales, by area



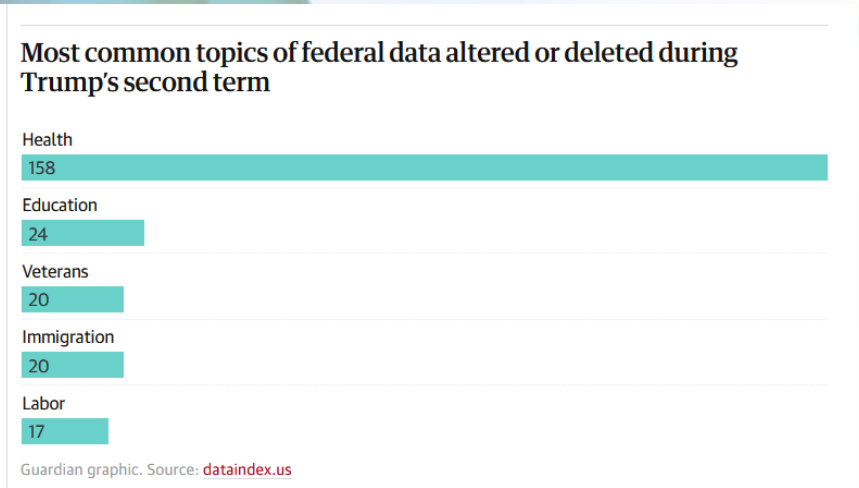
Source: National Bureau of Statistics
Jason French/WSJ

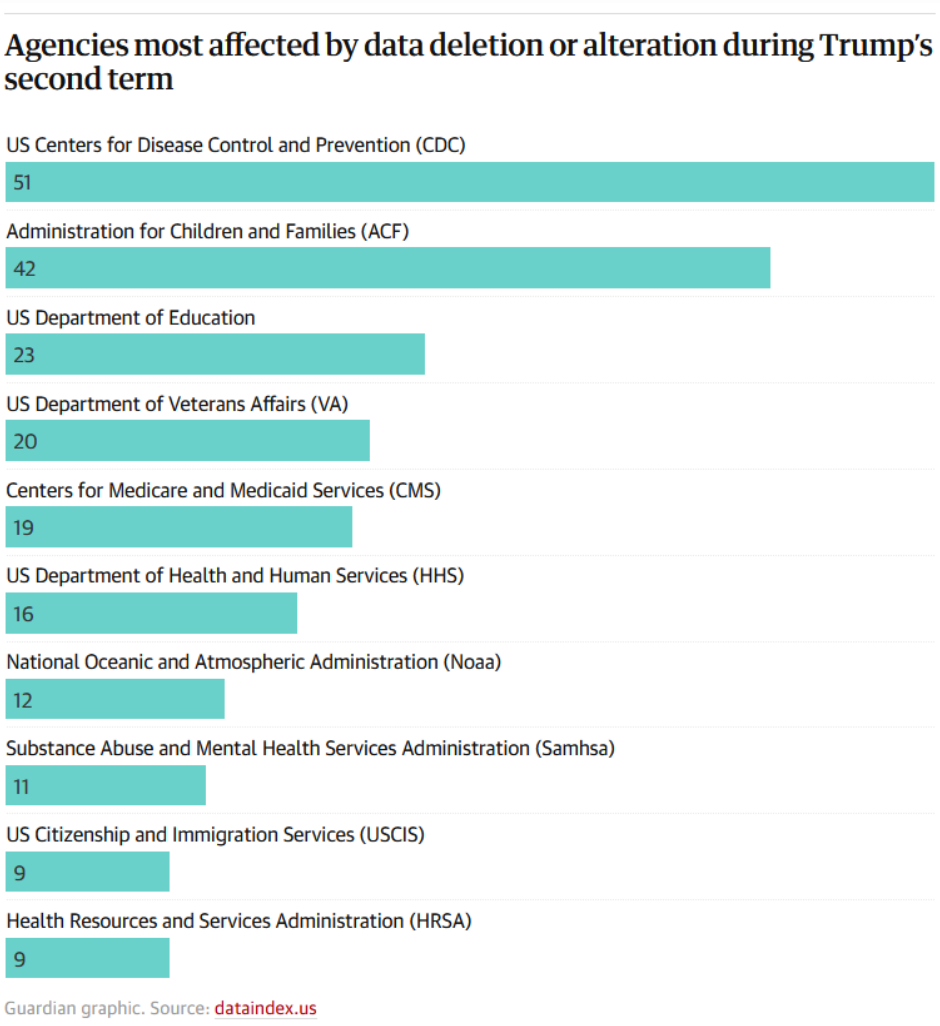
China – not alone with a similar problem….

- North Korean data
  - What can we use instead?
    - Kim (2022): The North Korean economy seen by satellite: Estimates of national performance, regional gaps based on nighttime light

- The most disappointing current case?
  - Freilich et al (2025): Disappearing Data at the U.S. Federal Government
    - *According to our analysis, between January 21 and February 11, 2025, the Centers for Disease Control and Prevention (CDC) removed 203 datasets (13% of its online datasets, a reduction from 1519 to 1316). The situation is dynamic; various datasets have been removed, restored, manipulated, or not posted as planned.*
      - *https://www.nejm.org/doi/full/10.1056/NEJMp2502567*
  - *Trump 2.0 has deleted or altered nearly 400 US datasets, endangering public health, education and more*
    - *https://www.theguardian.com/us-news/ng-interactive/2026/aug/18/trump-federal-data-deleted-altered*

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Source: https://www.theguardian.com/us-news/ng-interactive/2026/aug/18/trump-federal-data-deleted-altered

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Synthetic Control Method

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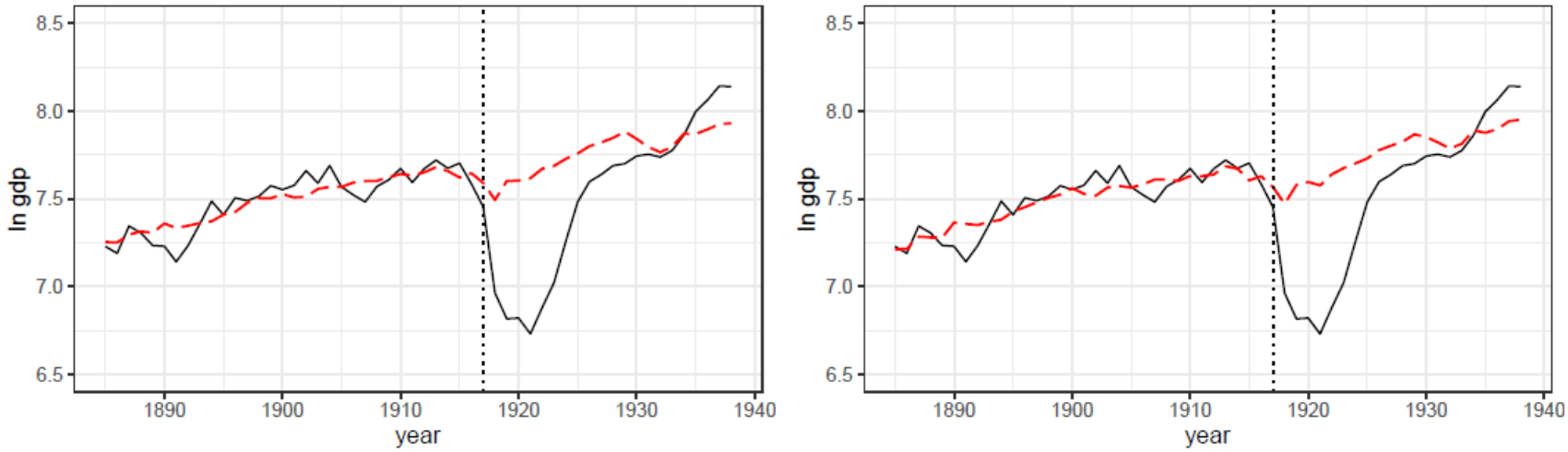
Korolev (2021): How Could Russia Have Developed without the Revolution of 1917?

- Constructs the counter-factual for Russia for 1917–1938 using several variations of the **synthetic control method**
  - Without the Revolution, Russia would probably have avoided the dramatic economic decline of the early 1920s and would have grown steadily throughout most of the 1920s.
  - Further results sensitive to specification
    - While most specifications suggest that the counterfactual Russia would have reached a lower level of economic development than the Soviet Union
      - Possible explanations:
        - Issues with the construction of the benchmark?

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Korolev (2021): How Could Russia Have Developed without the Revolution of 1917?

Figure 2: Actual and Counterfactual Log GDP per Capita, Synthetic Control



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Left: synthetic control method. Right: synthetic control method, demeaned.

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German Reunification: West Germany

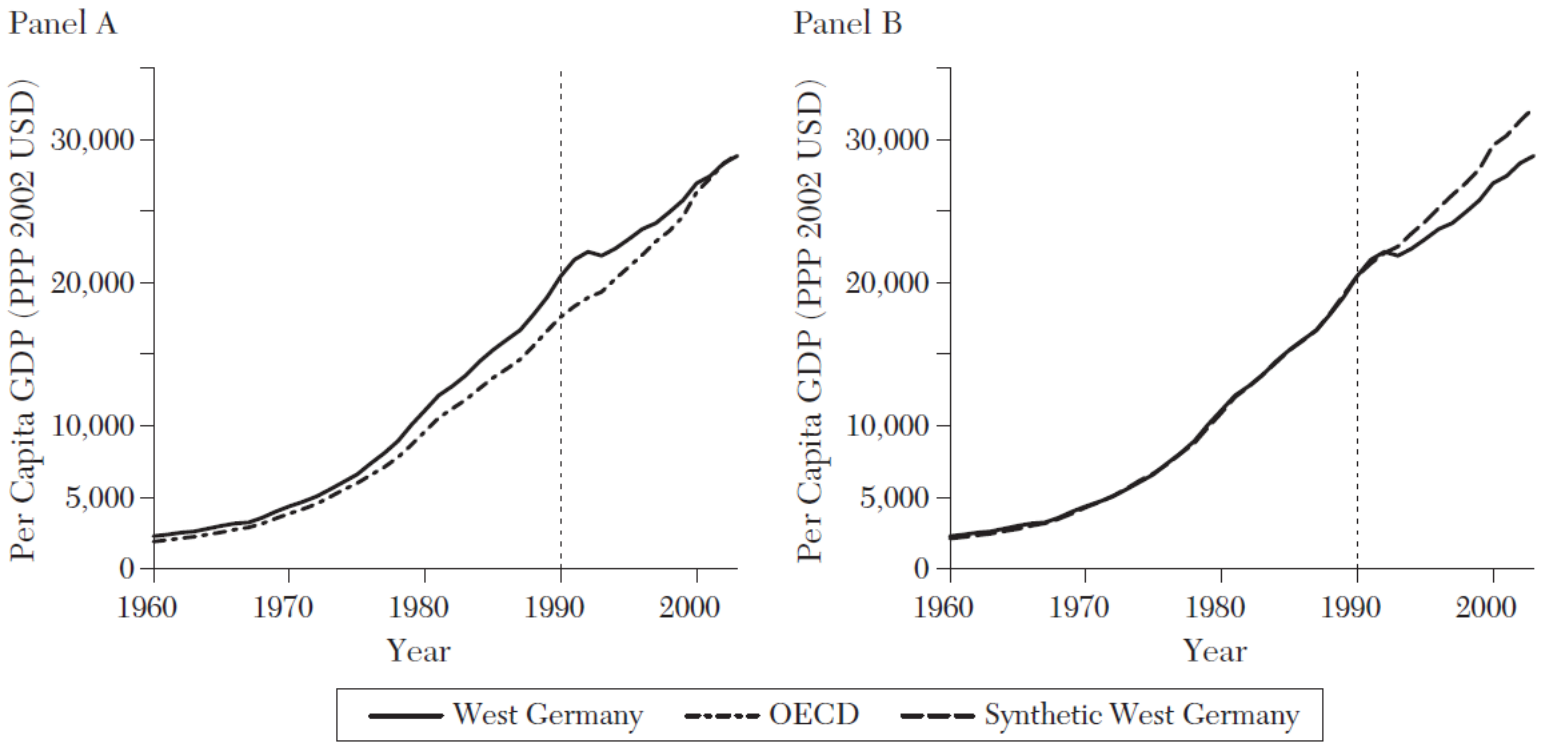


Figure 1. Synthetic Control Estimation in the German Reunification Example

Notes: Panel A compares the evolution of per capita GDP in West Germany to the evolution of per capita GDP for a simple average of OECD countries. In panel B the comparison is with a synthetic control calculated in the manner explained in subsection 3.2. See Abadie, Diamond, and Hainmueller (2015) for details.

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Other Examples of the Use of the SCM

- Abadie, Alberto. "**Using synthetic controls: Feasibility, data requirements, and methodological aspects**." *Journal of economic literature* 59.2 (2021): 391-425.
- Gilchrist, D., Emery, T., Garoupa, N., & Spruk, R. (2023). **Synthetic control method: A tool for comparative case studies in economic history**. *Journal of Economic Surveys*, 37, 409–445.
- Andreas Billmeier, Tommaso Nannicini; **Assessing Economic Liberalization Episodes: A Synthetic Control Approach.** The Review of Economics and Statistics 2013; 95 (3): 983–1001
- Kantorowicz, J., & Spruk, R. (2024). **Using synthetic control method to estimate the growth effects of economic liberalisation**: Evidence from transition economies. The World Economy, 47, 2332–2360.

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If you want to experiment…

- Ponne (2023): When Numbers Meet Stories – an introduction to the synthetic control method in R
  - https://www.r-bloggers.com/2023/07/when-numbers-meet-stories-an-introduction-to-the-synthetic-control-method-in-r/
  - Replicates and demonstrates Abadie's results for West Germany

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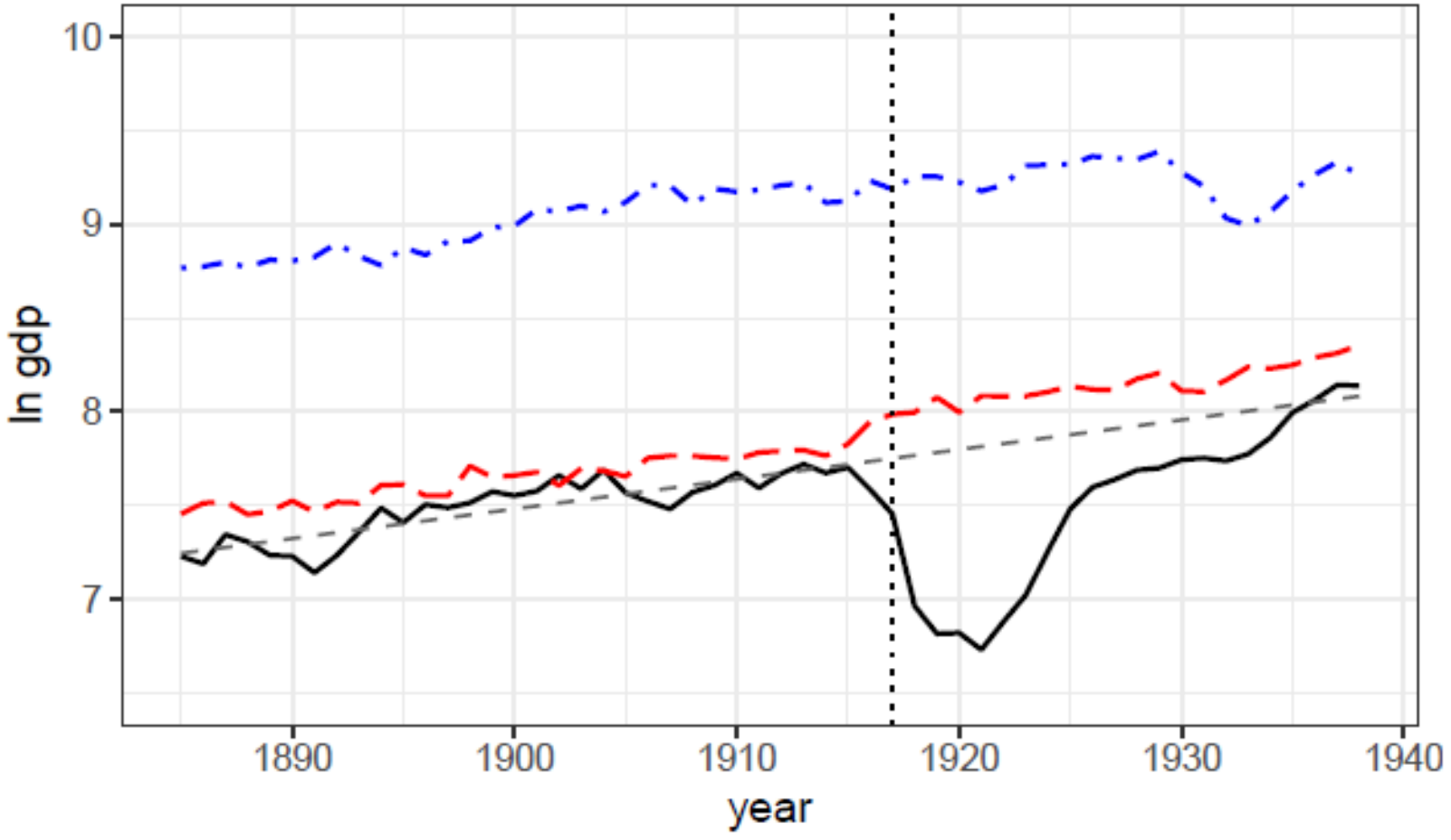
How does the SCM work?

- Policy effect = Observed outcome – Counterfactual outcome
- **Countefactual outcome:**
  - "We need an estimate of the country's (region's) alternative history."
  - A synthetic unit (country)
- Synthetic unit
  - Other countries provide possible comparisons.
  - Each may resemble our city in some ways.
  - A **weighted combination** may reproduce its history more closely.
    - How to get the weights?

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Korolev (2021): How Could Russia Have Developed without the Revolution of 1917?

Figure 1: Log GDP per Capita in Russia, Japan, and USA



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Abadie's Example: West Germany

TABLE 2
SYNTHETIC CONTROL WEIGHTS FOR WEST GERMANY

Australia	—
Austria	0.42
Belgium	—
Denmark	—
France	—
Greece	—
Italy	—
Japan	0.16
Netherlands	0.09
New Zealand	—
Norway	—
Portugal	—
Spain	—
Switzerland	0.11
United Kingdom	—
United States	0.22

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How do we choose the weights?

- Start with suitable untreated countries: **the donor pool.**
  - Countries in the donor pool should not have been hit by the "treatment" (or influenced by it indirectly – e.g. spillovers)
- Choose weights to match the treated city's outcomes before the policy.
  - In the standard method, weights are nonnegative and sum to one.
  - The algorithm searches for a combination that resembles the country before treatment.
    - Of course, we do not choose weights to create a desired result afterward.

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SCM: Problems

- In fact, it is often quite unlikely that the policy/event should affect only the treated country (without any effects on the donor pool)
  - Think of BREXIT
- With countries (more than with other units) and annual data it is difficult to separate one event/treatment
- Some shocks might be expected (people anticipate and change their behaviour even before the policy starts)
- Implications:
  - A close pretreatment fit can signal good comparability, but cannot prove that the counterfactual is correct.

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SCM: Tests

- Could a similar gap appear without treatment?
- Placebo tests
  - Pretend an untreated country received the policy.
    - Construct a synthetic control for that city.
    - Repeat across donor countries.
    - Compare their gaps with the treated country's gap, accounting for pretreatment fit.
- Similarly, we can test how the counterfactual looks like if we choose different treatment dates.

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Abadie's Example: W. Germany

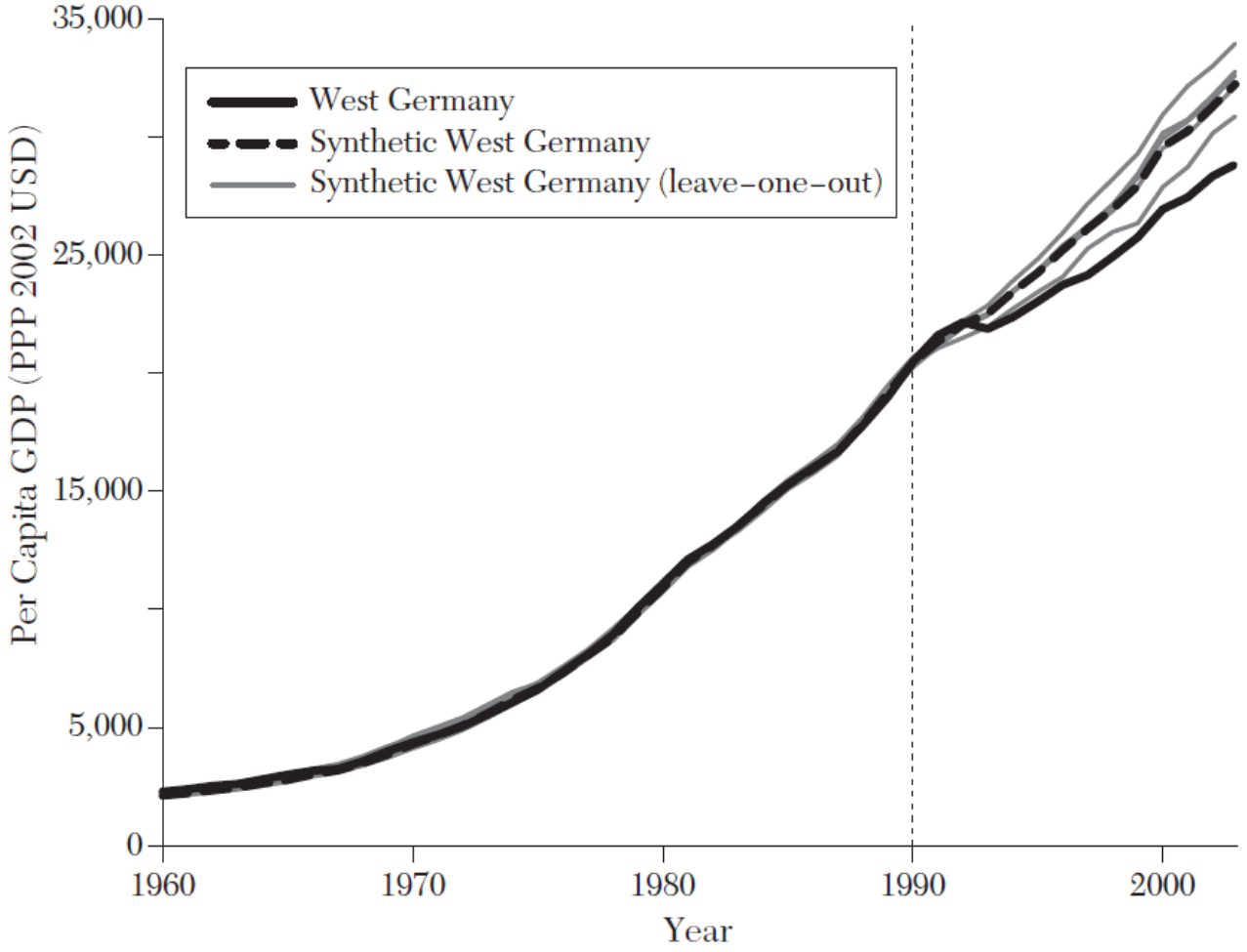


Figure 4. Leave-one-out Estimates of the Effect of the 1990 German Reunification

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Summary: Usefulness of the SCM

- Promising settings:
  - One or a few treated cities, regions, countries, or organizations.
  - Many observations before treatment.
  - Rich data on various features of the countries.
  - A credible set of untreated comparisons.
- Difficult settings:
  - The treated unit has a history no donor combination can reproduce.
  - Treatment affects the donor pool.
  - The intervention coincides with other distinctive shocks.
- SCM provides a transparent estimate of the missing untreated outcome.
  - But: the causal interpretation still depends on whether that comparison is credible.

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Seminar Session

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Reading Assignment for This Session (October 9th)

- Roland – chapter 7: Institutions and Economic Development
  - What are institutions? What types does the author mention?
  - What is the hold-up problem?
  - How would you explain the commitment problem?
  - Why can imperfect institutions persist?

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What Are Institutions?

- Institutions ≠ organizations!!!

- D. North (1990): "Institutions are the **rules of the game** in a society or, more formally, are the **humanly devised constraints that shape human interaction**."
- Three key features implied:
  1. they are "humanly devised," which contrasts with other potential fundamental causes, like geographic factors, which are outside human control;
  2. they are "**the rules of the game**" setting "constraints" on human behavior;
  3. their major effect will be through incentives
- Forms of institutions
  - Written laws and unwritten social norms.
- Role of institutions:
  - Effective institutions reduce uncertainty, cheating, and transaction costs

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Roland: Types of Institutions

Type	What it governs	Examples
Formal political institutions	Selection of political leaders and the distribution of political powers	Electoral rules, separation of powers
Formal legal institutions	Legally codified rights, obligations, and transactions	Property rights, contract rules
Informal institutions	Accepted and expected behavior without legal codification	Honesty, hospitality, customs and conventions

- **Formal institutions:** written rules with officially specified enforcement.
- **Informal institutions:** social norms enforced through peer pressure, guilt, and moral obligation.

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What is the hold-up problem?

- Situation:
  - One party makes an investment specific to a business relationship.
  - After the investment, the other party demands less favorable terms.
  - The investor has limited alternatives because the investment is already sunk.
  - **Anticipating this risk can discourage investment.**
- Key issues:
  - Asset specificity, sunk costs, and opportunistic behaviour
- Solution:
  - Enforceable contracts (and cheap enough litigation)

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What is the commitment problem?

- Situation:
  - A party has an incentive to break an earlier promise.
  - It often arises when transactions occur in sequence.
  - A promise that initially serves someone's interests may become unattractive later: **time inconsistency**.
- Solution
  - Legally enforceable contracts and penalties change the incentives to renege
  - Credible enforcement makes commitments more reliable.
- Hold-up or commitment?
  - Hold-up is a specific commitment problem involving a relationship-specific investment.
  - The broader commitment problem requires no investment or sunk cost.

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Why can inefficient institutions persist?

- Useful institutions do not emerge automatically.
- Reform requires collective action.
- Individuals may avoid the costs of reform while hoping to benefit from others' efforts.
  - Free riding
  - Olson: Small groups with a large stake can organize more effectively than large groups with a small individual stake.
- Existing institutions may obstruct people's ability to organize.
- Weak democratic institutions make change harder.

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Functionalist fallacy

- Functionalist fallacy:
  - assuming that an institution's useful function explains its existence, or that society will automatically develop the institutions it needs.
  - Institutions can arise for other reasons and remain inefficient

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Reading Assignment for This Session (October 9th)

- Acemoglu & Robinson – Why Nations Fail, chapter 1: So Close and Yet So Different
  - What seem to be the main differences between the „Nogales“
  - What was the encomienda and how did it work?
  - Was encomienda or a similar system used in Jamestown?
  - What kind of important historical differences between Mexico and USA are explained in the text?
  - How did Carlos Slim become rich?
  - How do the authors understand institutions?

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Two “Nogales”

Dimension	Nogales, Arizona, USA	Nogales, Sonora, Mexico
Household income	About $30,000 annually	About one-third of the Arizona level
Education	Most adults have completed high school	Many adults have not completed high school
Health	Higher life expectancy; better access to healthcare	Lower life expectancy; higher infant mortality
Public services	More reliable infrastructure and services	Poorer roads and fewer public amenities
Business environment	Greater security and protection of investments	More crime, bureaucracy, and corruption
Political accountability	Residents can replace elected representatives	Dominated by one party for many years (PRI), but a more recent experience of competitive democracy

What was encomienda?

- The Spanish colonial authorities granted an encomendero rights to receive tribute and labour services from a designated Indigenous population.
- In return, the encomendero was charged with converting that population to Christianity.
- In practice, the system enabled coercion and exploitation + It transferred resources and labour to a privileged colonial elite.

- The central feature was control over people's tribute and labor obligations.

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Did Jamestown use encomienda or a similar system?

- Jamestown did not establish the Spanish encomienda system.
- Its founders initially attempted a similar strategy of extracting resources and labor from Indigenous people.
- When that failed, the Virginia Company imposed coercive labor rules on English settlers.
- Eventually, it shifted toward land ownership and incentives to work.
- Why?

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Institutions in Acemoglu & Robinson?

Economic institutions	Political institutions
Shape opportunities to study, work, save, invest, and innovate	Determine who exercises power and how that power is constrained
Govern property rights, market entry, competition, and access to finance	Determine citizens' ability to influence and hold leaders accountable

Both can be inclusive or extractive, what is the difference?

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Inclusive and Extractive Economic Institutions

Dimension	Inclusive institutions	Extractive institutions
Main characteristic	Enable broad participation in economic activity	Channel income and wealth toward a privileged group
Property rights	Protect rights across broad sections of society	**Protection is selective**; others face insecurity or expropriation
Opportunities	Allow people to choose occupations, develop skills, and start businesses	**Restrict opportunities through coercion**, discrimination, or entry barriers
Competition	Permit entry and competition	Protect monopolies and **politically connected businesses**
Incentives	Encourage investment, innovation, and productive activity	Encourage obtaining privileges and extracting resources from others

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Inclusive and Extractive Political Institutions

Dimension	Inclusive institutions	Extractive institutions
Distribution of power	Power is distributed broadly: **pluralism**	Power is concentrated in a narrow group
Constraints	Effective limits on rulers and accountability	**Weak constraints on those controlling power**
State capacity	Sufficient central authority to enforce rules and maintain order	May involve either a powerful authoritarian state or an ineffective state
Economic consequences	Support broad opportunities and limit privileged groups' ability to dominate	Enable elites to create and **preserve economic privileges**

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Theoretical Framework

- Authors: Acemoglu & Robinson (2006)
  - Synthetic work
- Focus: development, institutions - linkages and causality
- Two key building blocks: economic and political institutions
  - Political institutions as main driving force (+ resource distribution)

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Theoretical Framework

Economic Institutions

- Shape incentives of key actors
- Influence on investments (physical, human capital or technology), production organization
- Impact on distribution
- Economic x cultural or geographical factors (Acemoglu, et al. 2002)

$$ec.institutions_t \Rightarrow \begin{cases} ec.performance_t \\ distr.of\ resources_{t+1} \end{cases}$$

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Theoretical Framework

Endogeneity

- **EI** as collective choices of society → conflict of interest
- Resulting EI depends on the political power of the proponents
  - Efficiency second-rate factor

$$political\ power_t \Rightarrow ec.institutions_t$$

Commitment problem?

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Theoretical Framework

Endogeneity of "de jure" Pol. Power

- Political power
  - De jure (institutional)
  - De facto
- Political institutions determine incentives
- E.g. monarchy x constitutional monarchy

$$pol.institutions_t \Rightarrow de\ jure\ pol.power_t$$

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Theoretical Framework

Endogeneity of "de facto" political power

- Other forms of pol. power: revolt, private army, economically costly protests
- Source of de facto political power
  - Ability to solve collective action problem
  - Economic resources
    - Main focus

$$resource\ dist._t \Rightarrow de\ facto\ pol.power_t$$

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Theoretical Framework

Endogeneity of Political Institutions

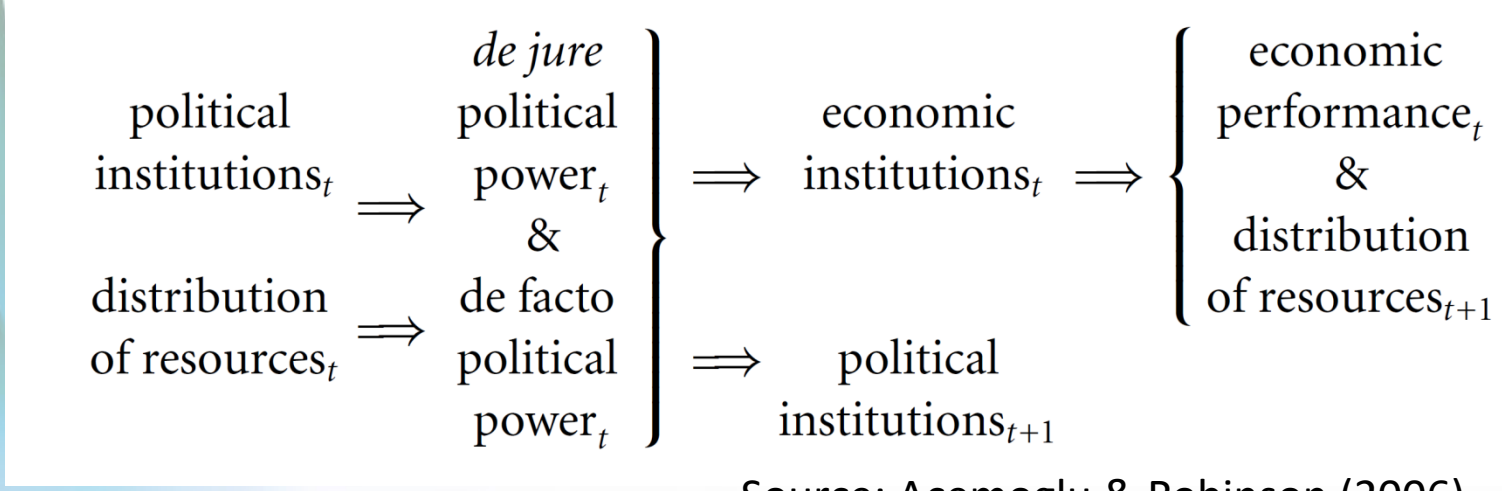
- PI also collective choices
- Distribution of political power as a key determinant of state evolution → persistence
- Sufficient de facto PP can cause a change

$$political\ power_t \Rightarrow pol.institutions_{t+1}$$

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Theoretical Framework

Summary



Source: Acemoglu & Robinson (2006)

- Sources of persistence → wealth disparity reproduction
  - PI durable
  - Wealth/resources of leading groups

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References

- Acemoglu & Robinson (2006): Paths of Economic and Political Development
- Acemoglu & Robinson – Why Nations Fail, chapter 1: So Close and Yet So Different
- Abadie, Alberto. "Using synthetic controls: Feasibility, data requirements, and methodological aspects." *Journal of economic literature* 59.2 (2021): 391-425.
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- Roland – chapter 7: Institutions and Economic Development

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